

■ CricStar Discord Bot

Complete Reference Guide

Everything you need to know about the bot — commands, file locations, how cards are created, how spawning works, and where to change things. Use this document alongside any AI assistant for quick help.

- Project root: /home/runner/workspace/
- Language: Python 3.12 | Framework: discord.py 2.x + Django
- Database: PostgreSQL (via DATABASE_URL env var)
- Bot token env var: DISCORD_BOT_TOKEN
- Start command: python run.py

1. Project Architecture

CricStar is a **Django + discord.py** application. Django manages the database, the admin web panel, and all models. discord.py runs the actual Discord bot. Both share the same database and run in a single process.

Command / Trigger	What it does
run.py	Entry point. Runs DB migrations, initialises settings in DB, sets up special events, then starts the bot.
admin_panel/	Django project — settings, models, migrations, admin panel, management commands.
admin_panel/bd_models/	Core Django app — Ball, BallInstance, Special, Regime, Economy models and their migrations.
admin_panel/settings/	Django app — global bot settings model (token, bot name, collectible name, etc.).
admin_panel/media/	Uploaded card images, event configs (event_config.json), neon colours (neon_colors.json).
cricstar/	The discord.py bot source code — core bot class, packages (cogs), image generator.
cricstar/core/bot.py	Main CricStarBot class — loads packages, connects to Discord.
cricstar/core/image_generator/image_generator.py	Generates all card images using Pillow. Contains draw_card() and draw_premade_card().
cricstar/packages/	Each sub-folder is a discord.py Cog (command group).

How the bot starts (run.py flow)

- 1. Django migrations run → database schema is up to date.
- 2. Settings object is created/updated in DB with DISCORD_BOT_TOKEN from environment.
- 3. setup_specials management command runs → ensures T20 World Cup & IPL2026 exist in DB.
- 4. startbot management command executes → loads all Cogs and connects to Discord.

2. Database Models

Ball (a cricketer card template)

File: admin_panel/bd_models/models.py

Command / Trigger	What it does
country	Unique player name (display name on card)
health	BAT score shown on card
attack	BALL score shown on card
rarity	Internal spawn weight stored as 0.0–1.0 (spawn_chance / 100). NOT the badge rarity.
capacity_logic	JSON field — stores forced_special ID and badge_rarity (cosmetic).
capacity_name	Codename shown on card (e.g. KING KOHLI)
capacity_description	Description paragraph shown on card
wild_card / collection_card	Filename of card image in admin_panel/media/
enabled	If False, card won't spawn and won't show in completion list
spawnable	If False, card never spawns randomly (even if enabled). Set when spawn_chance=0.
tradeable	Whether card can be traded
catch_names	Semicolon-separated alternate names players can type to catch
regime	FK → Regime (background/team group)
economy	FK → Economy (icon shown on card)

BallInstance (a specific card owned by a player)

File: admin_panel/bd_models/models.py

Command / Trigger	What it does
player	FK → Player (the Discord user who owns it)
ball	FK → Ball (which card type this is)
attack_bonus / health_bonus	Random ±% stat bonuses applied when caught (shown as +0%/–5%)
special	FK → Special event (if this instance is a special edition)
tradeable	Overrides ball.tradeable for this specific instance
trade_player	Set during a trade, tracks who offered it
deleted	Soft-delete flag — deleted instances stay in DB
locked	If True, instance cannot be traded
favorite	If True, shown with heart emoji in collection
shiny	Special shiny variant

Special (an event background)

Command / Trigger	What it does
name	Event name (e.g. IPL2026)

catch_phrase	Message shown when a special card is caught
start_date	When the event starts. MUST be set for the event to be randomly applied.
end_date	When the event ends. If blank, runs forever once started.
rarity	Weight 0.0–1.0 — chance this event's background is applied to a spawn.
background	Event background image file in admin_panel/media/
hidden	Hides the event from user-facing commands
tradeable	Whether cards with this special can be traded

■■ A Special with NO start_date is NEVER randomly applied. Set a start_date to activate it.

3. How Spawning Works

File: `cricstar/packages/cricketers/cricketer.py` and
`cricstar/packages/cricketers/countryball.py`

Step-by-step spawn flow

- 1. A message is sent in a configured spawn channel.
- 2. The SpawnManager checks if the cooldown period has passed.
- 3. A random Ball is selected weighted by `spawn_weight(rarity)`: if $\text{rarity} \leq 1.0$, $\text{weight} = \text{rarity}$; if $\text{rarity} > 1.0$, $\text{weight} = \text{rarity} + 25$.
- 4. `get_random_special()` is called — picks an active Special event (if any) based on its rarity weight. A Special is only 'active' if its `start_date` is set and in the past.
- 5. The card image is displayed in the channel with a 'Catch me!' button.
- 6. A player clicks the button and types the player's name.
- 7. A BallInstance is created for that player with random attack/health bonuses.
- 8. The catch message is displayed (e.g. 'You caught Virat Kohli! (#1A, +5%/−3%)').

Spawn weight explained

- `rarity` stored in DB = $\text{spawn_chance} \div 100$ (e.g. 90% $\rightarrow$ 0.9)
- `spawn_weight` for $\text{rarity} \leq 1.0 \rightarrow \text{weight} = \text{rarity}$ (higher rarity = spawns more often)
- `spawn_weight` for $\text{rarity} > 1.0 \rightarrow \text{weight} = \text{rarity} + 25$ (legacy support)
- Badge rarity (e.g. 20.0) is cosmetic only — stored in `capacity_logic["badge_rarity"]`, shown in `/csrarity`, does NOT affect spawn frequency.

4. Creating & Editing Cards

/cardmaker command

File: cricstar/packages/admin/cog.py (function: cardmaker)

Owner-only prefix command. Creates a new cricketer card and adds it to the database.

Command / Trigger	What it does
player_name	Unique DB/file identifier. Used for file naming.
display_name	Name shown on card header. If blank, player_name is used.
codename	Bold codename shown below image (e.g. KING KOHLI)
description	Description text (max ~256 chars shown on card)
bat_score	BAT / health stat number
ball_score	BALL / attack stat number
rarity	Badge display number shown on card (cosmetic, e.g. 20.0). Stored in capacity_logic.
spawn_chance	0–100%. How often card spawns. Stored as rarity = spawn_chance/100 in DB. 0 = never spawns.
artwork_author	Credit name shown at bottom of card
background	Preset name or image URL for card background
foreground	Player image URL or preset name (auto-saved for reuse)
logo_url	Optional team/event logo shown top-right of card
event	Assign to a special event — card ALWAYS spawns with that event's background
tradeable	Whether card can be traded (default: True)
catch_name	Alt names to catch; semicolon-separated (e.g. virat;vk;king)

/editcard command

Owner-only prefix command. Updates an existing card. All parameters are optional.

Same parameters as /cardmaker. Provide only the fields you want to change. Use the **regen** flag to regenerate the card image after making changes.

■■ When you update rarity via /editcard, the badge_rarity in capacity_logic is also updated automatically.

Where card images are stored

- admin_panel/media/ — all card image files live here
- File naming: premade_{playername}.webp (generated by /cardmaker)
- Image size: 1428 × 2000 pixels (portrait)
- Image format: WEBP (lossy, small file size)
- Foreground presets: saved inside the media folder by player slug name

5. Card Image Layout

File: cricstar/core/image_generator/image_gen.py (function: draw_premade_card)

Command / Trigger	What it does
Top bar (Y ≈ 140)	Player name (left, white) + Badge Rarity number (right, gold). Font: Nulshock-Bold 122px.
Image frame	Foreground player image. Starts at Y = TOP_BAR_H + 137 = 267px. Height = 716px.
Codename row	CODENAME: ... in white bold. Below the image frame.
Description	Word-wrapped description text below codename.
Logo (top-right of text)	145×145px team/event logo, right-aligned next to codename.
Bottom stats bar	BAT icon + number (left) and BALL icon + number (right). Height = 180px.
Credits (bottom-left)	'Created by El Laggron' + 'Artwork: {author}' in small text.

To adjust positions — edit these constants in image_gen.py

Command / Trigger	What it does
NAME_RARITY_Y = TOP_BAR_H // 2 + 7	Vertical position of player name & rarity text
FRAME_Y = TOP_BAR_H + 137	Top of the player image frame
CODENAME_Y = INFO_Y + 72	Vertical position of CODENAME: text
DESC_Y = CODENAME_Y + 104	Vertical position of description text start
BAR_Y = CARD_H - BAR_H (1820)	Top of the bottom stats bar
STAT_CY = BAR_Y + BAR_H // 2 - 18	Vertical centre of stat icons and numbers

6. All Bot Commands

■ Admin Commands (owner/staff only)

These are prefix commands (not slash). Owner = whoever is set as bot owner in Discord Developer Portal.

Command / Trigger	What it does
<code>.admin cardmaker</code>	Create a new cricketer card (full card generation)
<code>.admin editcard</code>	Edit an existing card (image, stats, name, rarity, etc.)
<code>.admin rarity</code>	List all cricketers ranked by spawn rarity
<code>.admin status</code>	Show bot status (guilds, cached cards, etc.)
<code>.admin trade_lockdown</code>	Lock all trades globally with a reason message
<code>.admin cooldown</code>	Show spawn cooldown info for a guild
<code>.admin guilds</code>	List guilds a user shares with the bot
<code>.admin syncslash</code>	Sync slash commands to Discord
<code>.admin setspawnimg</code>	Set a spawn image for a ball without regenerating the card
<code>.admin createevent</code>	Create a new special event
<code>.admin removecard</code>	Delete a card from the database
<code>.admin impersonate</code>	Act as another user for testing
<code>.admin admincolorset</code>	Set the neon glow colour for a player card
<code>.admin addemoji</code>	Assign an emoji to a card
<code>.admin imageadd</code>	Add/replace card image from file upload

■ Slash Commands (owner only)

Command / Trigger	What it does
<code>/csrarity</code>	Show all cricketers ranked by badge rarity (rarest first)
<code>/csresetcooldown</code>	Reset spawn cooldown for all guilds immediately
<code>/adminsynchron</code>	Sync slash commands globally
<code>/csdeletepath</code>	Delete a card image file by path
<code>/cscleanup</code>	Delete BallInstances from trades older than 15 days
<code>/csmultispawn</code>	Manually trigger a multi-card spawn

■ User Slash Commands

Command / Trigger	What it does
<code>/cricstar list</code>	Show your cricketer collection
<code>/cricstar completion</code>	Show your completion percentage
<code>/cricstar info</code>	Info about a specific cricketer card
<code>/cricstar last</code>	Show your most recently caught card
<code>/cricstar favorite</code>	Mark/unmark a card as favourite

<code>/cricstar give</code>	Give a card to another player
<code>/cricstar count</code>	Count how many of a card you have
<code>/cricstar duplicate</code>	Show your duplicate cards
<code>/cricstar compare</code>	Compare two players' cards
<code>/cricstar collection</code>	Show full collection of a player
<code>/csdaily</code>	Claim your daily free cricketer card (once per day)
<code>/csweekly</code>	Claim your weekly free cricketer card (once per week)
<code>/trade begin</code>	Start a trade with another player
<code>/trade add</code>	Add a card to an ongoing trade
<code>/trade remove</code>	Remove a card from an ongoing trade
<code>/trade history</code>	View your trade history
<code>/bet begin</code>	Start a bet with another player (winner takes all)
<code>/bet add</code>	Add a card to a bet
<code>/cricket upgrade</code>	Upgrade a cricketer's bat/ball stats
<code>/player settings</code>	Your personal privacy/trade settings
<code>/player info</code>	View your player profile
<code>/player friend add</code>	Add a friend
<code>/player friend remove</code>	Remove a friend
<code>/player friend list</code>	List your friends
<code>/player block add</code>	Block a user from trading with you
<code>/about</code>	Info about the bot
<code>/help</code>	Show command help list

■ ■ Server Config Commands (admin/owner)

Command / Trigger	What it does
<code>/setspawn channel</code>	Set the channel where cricketers spawn
<code>/setspawn remove</code>	Remove a spawn channel
<code>/setspawn enable</code>	Enable spawning in a channel
<code>/setspawn disable</code>	Disable spawning in a channel
<code>/setspawn cooldown</code>	Set the spawn cooldown (how often cards appear)
<code>/setspawn reset</code>	Reset the cooldown timer
<code>/setspawn status</code>	Show current spawn config for this server
<code>/config channel</code>	Set admin notification channel
<code>/config disable</code>	Disable the bot in this server
<code>/config status</code>	Show server config

7. Special Events (How to Run an Event)

Special events give cards a different background and a custom catch message. Two events already exist: **T20 World Cup** and **IPL2026**.

To start an event

- 1. Go to the Admin Panel → Special events.
- 2. Find the event (e.g. IPL2026).
- 3. Set a **start_date** to today (or any past date to activate immediately).
- 4. Optionally set an **end_date** — if left blank, the event runs forever.
- 5. The event's rarity (0.05 = 5%) is how often any random spawn gets the event background.
- 6. Restart the bot for the specials cache to refresh — or wait for the next cache reload.

Forced event on a specific card

When creating a card with `/cardmaker`, set the event parameter to the event name. That card will **always** spawn with that event's background, regardless of whether the event is currently active.

■■ Event background images are stored in `admin_panel/media/` and referenced in the Special model's background field.

8. Rarity — Two Different Things

There are TWO separate rarity values for each card. They are independent of each other.

1■■■ Badge Rarity (cosmetic) — the number shown on the card badge (e.g. 20.0).

Set with the `rarity` parameter in `/cardmaker`.

Stored in: `capacity_logic["badge_rarity"]`

Displayed by: `/csrarity` command

2■■■ Spawn Rarity (functional) — controls how often the card spawns.

Set with `spawn_chance` parameter (0–100%) in `/cardmaker`.

Stored in DB as: `ball.rarity = spawn_chance / 100`

Example: `spawn_chance=90` → `ball.rarity=0.9` → spawns very frequently.

Rarity values quick reference

Command / Trigger	What it does
<code>spawn_chance = 0</code>	Card never spawns randomly. <code>spawnable=False</code> .
<code>spawn_chance = 1-10</code>	Rare card. Spawns infrequently.
<code>spawn_chance = 50</code>	Common. Spawns often.
<code>spawn_chance = 90</code>	Very common. <code>Ball.rarity = 0.9</code>
<code>spawn_chance = 100</code>	Maximum. Always selected if in pool.
<code>badge rarity</code>	Any number. Only cosmetic — shown on card and in <code>/csrarity</code> . No effect on spawning.

9. Daily & Weekly Rewards

File: cricstar/packages/daily/cog.py

Command / Trigger	What it does
/csdaily	Gives a random enabled card once per day. Uses same spawn weight logic as regular spawns.
/csweekly	Gives a random enabled card once per week. Higher chance of rarer cards.

■ ■ Daily/weekly rewards can also be assigned special event backgrounds if an event is active.

10. Admin Panel (Web UI)

The Django admin panel lets you manage everything via a web browser. It runs on the same process as the bot.

Command / Trigger	What it does
<code>start_admin.sh</code>	Script to start the admin web server
<code>admin_panel/</code>	Django project root
<code>Settings</code> model	Bot token, bot name, collectible name, slash command name, spawn range, etc.
<code>Ball (Cricketer)</code> model	All card templates — create, edit, delete cards
<code>BallInstance</code> model	Every card owned by every player — view player inventories
<code>Special</code> model	Manage special events — set start/end dates to activate them
<code>Regime</code> model	Background groups (teams)
<code>Economy</code> model	Economy icons shown on cards
<code>Player</code> model	Discord player accounts and their settings

11. Key Files — Quick Reference

Command / Trigger	What it does
run.py	Startup script — entry point
cricstar/core/bot.py	Main bot class
cricstar/core/image_generator/image_generator.py	Card image generation (Pillow)
cricstar/packages/admin/cog.py	All owner/admin commands
cricstar/packages/cricketers/cricketers.py	Spawn logic, catch logic, special selection
cricstar/packages/cricketers/countryball.py	CountryBall helper class
cricstar/packages/collection/cog.py	/cricstar list, info, favorite, give, etc.
cricstar/packages/trade/cog.py	Trade system
cricstar/packages/bet/cog.py	Bet system
cricstar/packages/daily/cog.py	/csdaily and /csweekly
cricstar/packages/upgrade/cog.py	/cricket upgrade
cricstar/packages/players/cog.py	/player settings, friends, blocks
cricstar/packages/guildconfig/cog.py	Server config commands
cricstar/packages/setspawn/cog.py	Spawn channel setup
admin_panel/bd_models/models.py	All database models (Ball, BallInstance, Special, etc.)
admin_panel/settings/models.py	Global bot settings model
admin_panel/media/event_config.js	Legacy event config (use Special model instead)
admin_panel/media/neon_colors.json	Per-player neon glow colours
admin_panel/bd_models/management/commands/default_special_events.py	Creates default special events on startup
admin_panel/bd_models/management/commands/django_management.py	Django management command that launches the bot

12. Environment Variables

Command / Trigger	What it does
DISCORD_BOT_TOKEN	Your Discord bot token. Required.
DATABASE_URL	PostgreSQL connection string. Required.
DJANGO_SETTINGS_MODULE	Always set to: admin_panel.settings.cricstar
CRICSTARBOT_DB_URL	Alternative DB URL (falls back to DATABASE_URL)

Setting up on a VPS

- 1. Copy the full project folder to your VPS.
- 2. Install Python 3.12 and run: `uv sync` (installs all dependencies).
- 3. Create a PostgreSQL database and get its connection URL.
- 4. Set the three environment variables above.
- 5. Run: `python run.py`
- 6. To keep it running: use `systemd`, `pm2`, or `screen/tmux`.
- 7. For updates: pull new code with `git pull`, then restart the bot.

13. Common Tasks — Where to Make Changes

Change bot name / collectible name

Admin Panel → Settings → bot_name / collectible_name. Restart bot.

Change slash command prefix (default: /cricstar)

Admin Panel → Settings → cricstar_slash_name. Restart bot.

Add a new card

Use .admin cardmaker in Discord (owner only). All params in Section 4.

Edit an existing card

Use .admin editcard. Add regen=True to regenerate the image.

Change card image position (name, rarity, stats)

Edit constants in cricstar/core/image_generator/image_gen.py. Regenerate cards.

Start a special event

Admin Panel → Special → set start_date. Restart bot or wait for cache refresh.

Add a new special event

Use .admin createevent or create it in Admin Panel.

Change spawn frequency

Edit the card's spawn_chance via .admin editcard spawn_chance=XX.

Disable a card from spawning

Set spawn_chance=0 in editcard, or set enabled=False in Admin Panel.

Reset daily/weekly cooldown for a user

Edit the Player record directly in Admin Panel.

Reset spawn cooldown

Use /csresetcooldown (owner only).

Add alternate catch names

.admin editcard catch_name='name1;name2;name3'

Set neon glow colour for a card

.admin admincolorset playername R G B (e.g. 0 210 255 for cyan)

Change the catch button label

Admin Panel → Settings → catch_button_label

Change min/max spawn cooldown range

Admin Panel → Settings → spawn_chance_min / spawn_chance_max

Quick AI Prompting Tips

When using an AI assistant with this document, try prompts like:

- "I want to change the position of the rarity text on the card. Which file and which constant do I edit?"
- "How do I add a card that only spawns during a specific event?"
- "The /csrarity command is showing wrong values. What could be the cause?"
- "I want to make a card that is very rare (1% spawn chance) but shows badge rarity 99. How?"
- "How do I start the IPL2026 event so cards spawn with that background?"
- "Where is the daily reward logic and how do I change the cooldown?"

End of Document